Module 2
IAM in GCP

Module 2: IAM

Goals: Understand about IAM, service account

- Add principals
- Assign roles
- Primitive, predefined, custom roles
- Service account
- Access and manage service account

Module 2: IAM

Technical Definition

➢ With Cloud IAM you can grant granular access to specific GCP resource and prevent unwanted access to other resources. Cloud IAM lets you adopt the security principle of least privilege, so you grant only necessary access to your resources.

➢ Who Can do what on which resources?

○ Who = Member

○ Can Do What = Role

○ On Which Resources = Resources in GCP

Module 2: IAM

➢ **Who = Member = Identity**

○ Person Authenticate with Identity (email address)

○ Service Account -- Server/Application access account, not associated with any person.

➢ **Can do What = Role = Group of Permissions**

○ Permission : What operation is allowed on Resource

○ Permission Format : service.resource.verb

Ex: compute.instances.edit

○ Permissions are not directly assigned to a Member(Who)

○ Permissions are bundled in Roles - Roles further assign to Members

Multiple Roles can be assigned to Single Member

Module 2: IAM

➢ On Which Resources = Resources in GCP

- All of the component of GCP
- Including Organization, Folder, Projects and resources inside.

Ex: Compute Engine, App Engine, ML APIs, Data Services, Network etc.

➢ Roles types

- Primitive
- Predefined
- Custom

Module 2: IAM

➢ Primitive Roles

- Broad, original roles available in GCP
- Applied across entire environment
- Owner: Modify all resource and manage IAM and Billing
- Editor: Modify all resource and No access to IAM and Billing
- Viewer: Only view resource

Module 2: IAM

➢ Predefined Roles

○ More Granular, specific access not access the entire project

○ Applied to single service

Example: compute.storageAdmin - Allow Full control of Compute Engine storage resources.

➢ Custom Roles

○ Even more granular than Predefined Roles

○ Hand Made Group of Permissions

Module 2: IAM

➢ Predefined Roles

- More Granular, specific access not access the entire project
- Applied to single service

Example: compute.storageAdmin - Allow Full control of Compute Engine storage resources.

Module 2: IAM

➢ What is Service Account?

- Special Type of Account, Not Attached to User.
- Service account represent by email address-

  Ex: 1234-compute@developer.gserviceaccount.com
- Allow authentication between Application and GCP Services.
- By Default, GCE instances use Service account to access GCP services.

Module 2: IAM

➢ Types of Service Account.

➢ Google Managed

○ Represent different google services and automatically generate IAM Roles.

➢ User Managed

○ Created for/by you. Based on Enable APIs in Project

○ [project-number]-compute@developer.gserviceaccount.com

○ [project-id]@appspot.gserviceaccount.com

○ Service Account can also be assigned to Person Email(Member)

Module 2: IAM

➢ Service Account keys

- Service Account access managed by account keys.

○ Assume it like an account password

- Default Service account keys are managed by google and can’t be accessed and edited.

- Custom service account can use (custom) user managed Keys-

○ Google Maintain Public Copy but Private key is user Managed.

○ If user lose private key, Google cannot retrieve it.

○ Can Also Used Google Managed Keys.

Module 2: IAM

➢ Lab

- Create a service account using the Google Cloud console.
- Download the service account key file.
- Grant the service account the appropriate permissions to upload file to the GCP bucket.
- Use the service account key file to authenticate to Google Cloud APIs and resources.

Thank you